

Udhav Ulhas Gawandalkar

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EDUCATION

PhD	2019-24	Multiphase Systems, Process and Energy, Mechanical Engineering, Delft University of Technology, The Netherlands	
Graduate Studies	2017-19	Master of Science (MSc), Energy Flow and Process, Mechanical Engineering, Delft University of Technology, The Netherlands	8.43/10 (cum laude) Graduation Thesis: 9/10
Undergraduate studies	2011-15	Bachelor of Technology, Mechanical Engineering, National Institute of Technology Karnataka (NITK), Surathkal, India	8.50/10 Graduation Thesis: 10/10

RESEARCH

Postdoctoral Researcher, Institute of Fluid Dynamics, ETH Zurich, Switzerland (Dec '24- present)

Title: ***Investigation of gravity currents: saline and particle-laden currents using optical and X-ray imaging***

Supervisor: Prof. dr. Filippo Coletti

- leading gravity currents research at Coletti lab—set up of new lock-exchange flow facility and X-ray radiography laboratory to study dense particle-laden gravity currents and other multiphase flows
- Investigating compositional currents with PIV+PLIF, with emphasis on its propagation, mixing, entrainment, stratification
- Investigating particle-laden gravity current with emphasis on particle-settling, turbulence, entrainment

Researcher, Multiphase Systems, Process & Energy, Delft University of Technology, The Netherlands (Nov '23–May '24)

Title: ***Particle migration in neutrally-buoyant dense suspensions in pipes***

Supervisor: Prof. dr. ir. Christian Poelma

- Performed pressure drop measurements for different particle concentrations and Reynolds numbers
- Studied the flow development and core-peaking with concentration and velocity measurements using ultrasound imaging velocimetry and light-attenuation techniques

PhD Researcher, Multiphase Systems, Process & Energy, Delft University of Technology, The Netherlands (Nov '19 - Oct '24)

Title: ***The experimental investigation of hydrodynamic cavitation in an axisymmetric venturi***

Supervisor: Prof. dr. ir. Christian Poelma, Prof. dr. ir. Jerry Westerweel

- Performed tomographic imaging and velocimetry of near-wall re-entrant flow to unveil its role in periodic vapour cavity shedding
- Performed high-speed X-ray densitometry measurements to uncover the characteristics of bubbly shock waves triggering cloud cavitation
- Above studies were combined to study the transition of shedding mechanism in partial cavitation phenomenon

Visiting Researcher, University of Michigan, Ann Arbor, Michigan, USA (Sept '22 - Jan '23)

Title: ***Ventilated cavities in the wake of a two-dimensional wedge via time-resolved X-ray densitometry***

Supervisor: Dr. Harish Ganesh, Prof. dr. Steven L. Ceccio,

- Setup experiments in the water tunnel to study cavity closure and gas ejection mechanism in ventilated cavities
- This was achieved via inhouse high-speed X-ray densitometry system (void fraction measurements) and high-speed imaging.

Master Thesis Student, Laboratory for Aero and Hydrodynamics, TU Delft, The Netherlands (Dec '18 - Oct '19)

Title: ***The effect of mildly rapid strain on turbulent pipe flow***

Advisor: Dr. ir. Gerrit. E. Elsinga, Prof. dr. ir. Jerry Westerweel

- A moderately rapid strain producing axisymmetric contraction was designed and installed on long pipe facility at TU Delft
- The response of turbulent scales (large and small) to the applied strain is studied using high-resolution Particle Image Velocimetry (PIV).

Graduate Intern, Heat transfer and Fluid Dynamics Group, TNO, Leeghwaterstraat, Delft, The Netherlands (Jul – Oct '18)

Title: ***An experimental investigation of the stability of foams***

Advisor: Ir. Pejman S. Omrani, Prof. dr. ir. R.A.W.M. Henkes

- Performed number of experiments to study the effect of salinity and nanoparticles on foamer performance in deliquifying gas wells—Bickermann sparging test, Ross Mile test and Dynamic Surface Tension measurements.

Research Assistant, Multiphase flow, Indian Institute of Science, Bangalore (IISc), India.

(Jul'16 – Jun'17)

Title: Fundamental study on adsorption of water vapors by Silica Gel

Advisor: Prof. dr. Saptarishi Basu

- Measured uptake of water by silica gel particle using micro force sensor, shadowgraphy, and Micro-PIV
- Developed Tunable Diode Laser Absorption Spectroscopy (TDLAS) technique to estimate the concentration gradients of water vapor around silica gel.

Undergraduate Thesis, National Institute of Technology Karnataka (NITK), Surathkal, India

(Aug'14 –Jul'15)

Title: Numerical Simulation of coronary artery blood flow in stenotic condition

Advisor: Dr. Arun M. , Dr. Jasjit Suri

- Reconstructed actual geometry of coronary artery from MRI data of real stenotic patients with the aid of digital image processing for varying degree of stenosis.
- Investigated wall shear stress and oscillatory shear index at specific sites using commercial finite volume CFD solver.

SELECTED PUBLICATIONS

- ♦ Gawandalkar U.U., Coletti F. ***Propagation and dynamics of lock release gravity currents down a slope.*** (2026) (manuscript in preparation)
- ♦ Gawandalkar U.U., Lucido N., Jain P., Poelma C., Ceccio S., Ganesh H. ***Examination of ventilated cavity in the wake of two-dimensional bluff body using X-ray densitometry.*** (2025) (Journal of Fluid Mechanics)
- ♦ Gawandalkar U.U., Poelma C. ***The characteristics of bubbly shock waves in a cavitating axisymmetric venturi via time-resolved X-ray densitometry.*** Journal of Fluid Mechanics (2024).
- ♦ Gawandalkar U.U., Poelma C. ***The structure of near-wall re-entrant flow and its influence on cloud cavitation instability.*** Experiments in Fluids (2022).
- ♦ John, K., Jahangir, S., Gawandalkar U. U., Hogendoorn, W., Poelma, C., Grundmann, S., Bruschewski, M. ***Magnetic resonance velocimetry in high-speed turbulent flows: sources of measurement errors and a new approach for higher accuracy.*** Experiments in Fluids (2020).
- ♦ Mahalingam A., Gawandalkar U.U., Kini G., et al. ***Numerical analysis of the effect of turbulence transition on the hemodynamic parameters in human coronary arteries.*** Cardiovascular Diagnosis Therapy (2016).

SELECTED PRESENTATIONS

- ♦ Oral Presentation at 78th Annual Meeting of the Division of Fluid Dynamics, APS, Houston, USA (2025)
Title: ***Dynamics of particle-laden gravity currents down a slope***
- ♦ Invited Speaker at Radiographic Imaging and Tomography (RadIT), Optica, Seattle, Washington, USA (2025)
Title: ***X-Ray Densitometry of Multiphase Flows***
- ♦ Oral presentation at 12th International Conference on Multiphase flows, ICMF, Toulouse, France (2025)
Title: ***Quantifying velocity and concentration profiles in dense suspensions***
- ♦ Oral presentation at 12th International Conference on Multiphase flows, ICMF, Toulouse, France (2025)
Title: ***Ventilation hysteresis in ventilated cavities in the wake of a bluff body***
- ♦ Oral presentation at 35th Symposium on Naval Hydrodynamics (SNH) Nantes, France (2024)
Title: ***X-ray measurements of ventilated cavities in the wake of a 2-D bluff body and an axisymmetric cavitator***
- ♦ Oral presentation at 20th International Symposium on Flow Visualization ISFV-20, Delft, The Netherlands (2023)
Title: ***X-ray densitometry of ventilated cavities in the wake of a bluff body***
- ♦ Oral presentation at 20th International Symposium on Applications of Laser and Imaging Techniques to Fluid Mechanics, Lisbon, Portugal (2022)
Title: ***Turbulent pipe flow subjected to a mild straining***
- ♦ Oral presentation at 75th Annual Meeting of the Division of Fluid Dynamics, APS, Indianapolis, Indiana, USA (2022)
Title: ***The characteristics of bubbly shock waves in a cavitating venturi via time-resolved X-ray densitometry***

- ♦ Oral presentation at 14th International Symposium on Particle Image Velocimetry, Chicago, USA (ISPIV 2021)
Title: *Tomographic PIV measurement of a re-entrant jet in a cavitating venturi*
- ♦ Oral presentation at 11th International Symposium on Cavitation, Daejeon, Korea (CAV2021)
Title: *Dynamics of Re-entrant Jet in a Cavitating Venturi*

DATASETS

- ♦ Gawandalkar U.U., Lucido, N.A., Jain, P., Poelma, C., Ceccio, S.L., Ganesh, H. *Data underlying the examination of ventilated cavities in the wake of two-dimensional bluff body using time-resolved X-ray densitometry*, 4TU.ResearchData (2024).
- ♦ Gawandalkar U.U., Poelma C. *Data underlying the characteristics of bubbly shock waves in a cavitating axisymmetric venturi using time-resolved X-ray densitometry*, 4TU.ResearchData (2024).
- ♦ Gawandalkar U.U., Poelma C. *Data underlying the quantification of re-entrant jet in a cavitating axisymmetric venturi using time-resolved tomographic velocimetry*, 4TU.ResearchData (2024).

PATENTS

Laser Based System for Studying Adsorption Kinetics of Adsorbent and Adsorbate Pair

(Indian Patent No. TEMP/E-1/46072/2017- CHE; Filing Date: 15/12/2017) Application number: 201741045155

WORKSHOPS

- ♦ 28th CISM-IUTAM Summer School on "Gravity Currents in the Environment", coordinated by prof. Eckart Meiburg and prof. Benjamin Kneller, Udine, Italy, 2025
- ♦ Master of Flow Visualization at 20th International Symposium on Flow Visualization Delft, The Netherlands, 2023

REVIEWER ACTIVITY

Journal of Fluid Mechanics, International Journal of Multiphase Flow, Experiments in Fluids, Acta Mechanica, Chemical Engineering Science, Ocean Engineering

STUDENT SUPERVISION

- ♦ Jasper Hüchting, M.Sc. Mechanical Engineering, ETH Zurich, (Fall 2025)
- ♦ Àngel Monferrer Cervelló, M.Sc. Mechanical Engineering, ETH Zurich, (Spring 2026)

OTHER EXPERIENCES

- ♦ Teaching Assistant, Mechanical Engineering, TU Delft, The Netherlands (Feb '20– sep '21)
 - conducted weekly problem solving sessions for master students in the course 'Turbulence' taught by Prof. Westerweel
 - Designed assignment questions and corrected exams
- ♦ Teaching Assistant, Mechanical Engineering, TU Delft, The Netherlands (Feb - Apr '19)
 - Instructed and conducted practicum class for undergraduate student in the course 'Strength of Ships'
 - The students were introduced to Digital Image Correlation (DIC) technique to measure shear lag in a loaded plate
- ♦ Graduate Engineer Trainee, Hero Moto Corp Limited (HMCL), India (Jul '15 - Jul '16)
 - worked on several projects within Engine Plant, and served as the Shift In charge overseeing production line
 - Involved in creating training modules for the machine operators and delivering classroom training sessions

POSTITIONS OF RESPONSIBILITY

- ♦ Treasurer at 'PantaRhei', a dispute board for Laboratory for Aero and Hydrodynamics, TU Delft, The Netherlands
- ♦ The team head of 'Suspension and Steering' subsystem and the treasurer at NITK-BAJA (an ATV racing team)
- ♦ Organizing committee for 'Fluid Mechanics and Fluid Power (FMFP-2015)', Surathkal, Karnataka, India

SCHOLARSHIP

Recipient of Japan Student Service Organization (JASSO) scholarship, 2015